

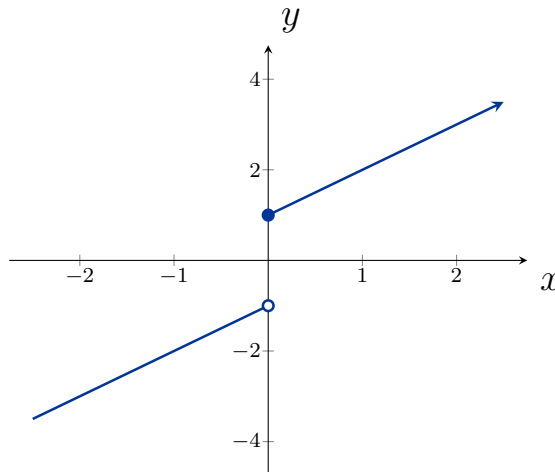
2.5: One-Sided Limits and Continuity

Consider the function

$$f(x) = \begin{cases} x - 1, & x < 0 \\ x + 1, & x \geq 0 \end{cases}$$

What is $\lim_{x \rightarrow 0} f(x)$?

$$\lim_{x \rightarrow 0} f(x) \text{ DNE}$$



Definition. (One-Sided Limits)

The function f has a **right-hand limit** L as x approaches a from the right, written

$$\lim_{x \rightarrow a^+} f(x) = L$$

if the values of $f(x)$ can be made as close to L as we please by taking x sufficiently close to (but not equal to) a and to the right of a .

The function f has a **left-hand limit** M as x approaches a from the left, written

$$\lim_{x \rightarrow a^-} f(x) = M$$

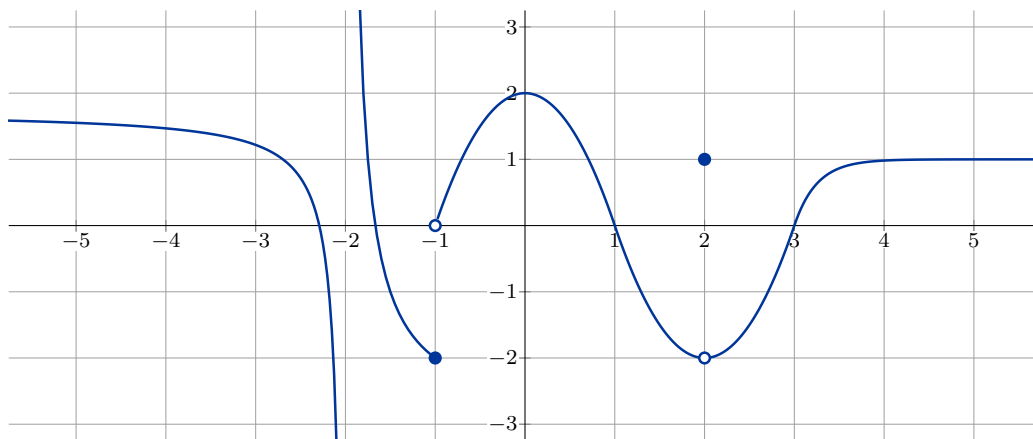
if the values of $f(x)$ can be made as close to L as we please by taking x sufficiently close to (but not equal to) a and to the left of a .

Theorem 3

Let f be a function that is defined for all values of x close to $x = a$ with the possible exception of a itself. Then

$$\lim_{x \rightarrow a} f(x) = L \quad \text{if and only if} \quad \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L$$

Example. Using the graph below, evaluate the following limits:



$$\lim_{x \rightarrow -2^-} f(x) = -\infty$$

$$\lim_{x \rightarrow -2^+} f(x) = \infty$$

$$\lim_{x \rightarrow -2} f(x) \text{ DNE}$$

$$\lim_{x \rightarrow -1^-} f(x) = -2$$

$$f(-1) = -2$$

$$\lim_{x \rightarrow -1^+} f(x) = 0$$

$$\lim_{x \rightarrow -1} f(x) \text{ DNE}$$

$$\lim_{x \rightarrow 1} f(x) = 0$$

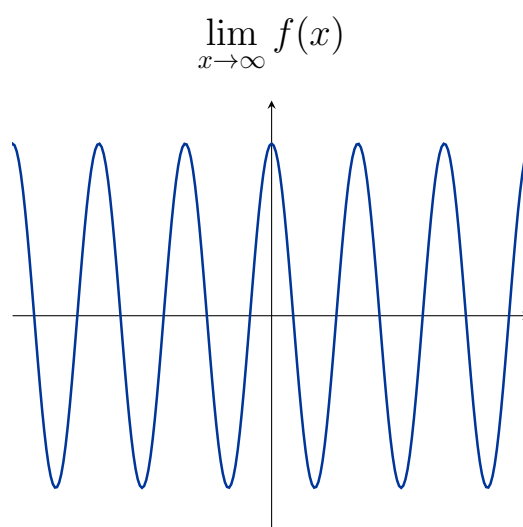
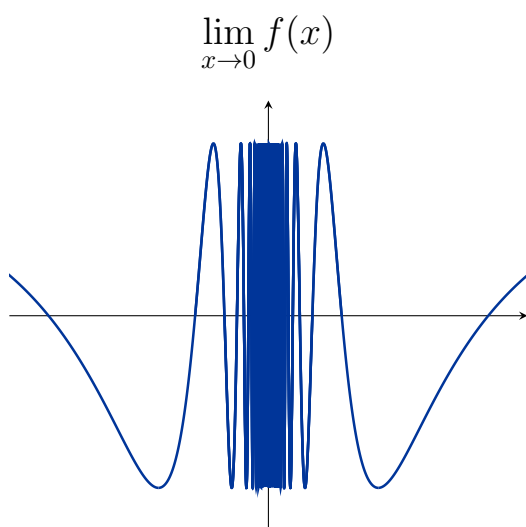
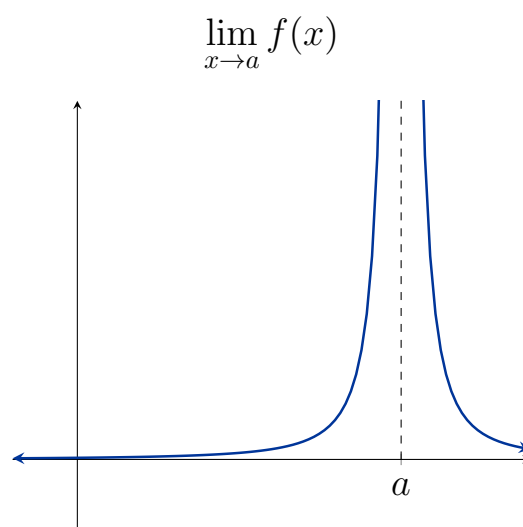
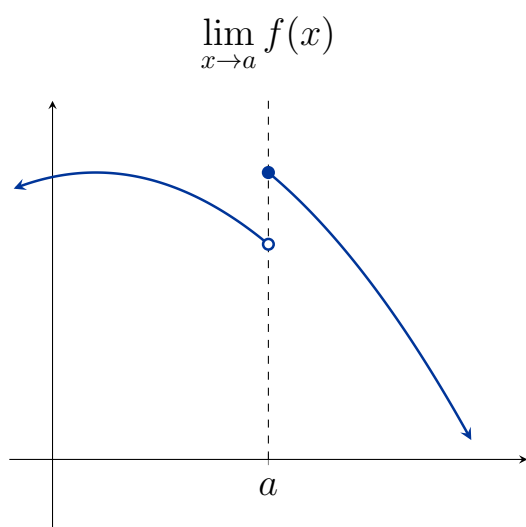
$$\lim_{x \rightarrow 2} f(x) = -2$$

$$f(2) = 1$$

$$\lim_{x \rightarrow \infty} f(x) = 1$$

Below are examples where the limit does not exist:

[Graph](#)



Definition. (Continuity of a Function at a Number)

A function f is **continuous** at a if $\lim_{x \rightarrow a} f(x) = f(a)$.

Continuity Checklist:

In order for f to be continuous at a , the following three conditions must hold:

1. $f(a)$ is defined (a is in the domain of f),
2. $\lim_{x \rightarrow a} f(x)$ exists,
3. $\lim_{x \rightarrow a} f(x) = f(a)$ (the value of f equals the limit of f at a).

Example. Determine the values of x for which the following functions are continuous:

$$f(x) = 3x^3 + 2x^2 - x + 10$$

Continuous
on $(-\infty, \infty)$

$$g(x) = \frac{8x^{10} - 4x + 1}{x^2 + 1}$$

$x^2 + 1 \neq 0$
 $x^2 \neq -1$ $\Rightarrow$ Continuous
on $(-\infty, \infty)$

$$h(x) = \frac{4x^3 - 3x^2 + 1}{x^2 - 3x + 1}$$

$x^2 - 3x + 1 \neq 0$
 $x \neq \frac{3 \pm \sqrt{(-3)^2 - 4(1)(1)}}{2} = \frac{3 \pm \sqrt{5}}{2}$
 $\Rightarrow$ Continuous
on $(-\infty, \frac{3-\sqrt{5}}{2}) \cup (\frac{3-\sqrt{5}}{2}, \frac{3+\sqrt{5}}{2}) \cup (\frac{3+\sqrt{5}}{2}, \infty)$

Example. Determine whether the following are continuous at a :

$$f(x) = x^2 + \sqrt{7-x}, \quad a = 4 \quad \checkmark$$

$$f(4) = 16 + \sqrt{3}$$

$$\lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^-} f(x) = \lim_{x \rightarrow 4^+} f(x) = \lim_{x \rightarrow 4} f(x) = 16 + \sqrt{3}$$

$$g(x) = \frac{1}{x-3}, \quad a = 3 \quad \times$$

$$g(3) \text{ DNE}$$

$$h(x) = \begin{cases} \frac{x^2+x}{x+1}, & x \neq -1 \\ 0, & x = -1 \end{cases}, \quad a = -1 \quad \times$$

$$\begin{aligned} \lim_{x \rightarrow -1} h(x) &= \lim_{x \rightarrow -1} \frac{x^2+x}{x+1} \\ &= \lim_{x \rightarrow -1} \frac{x(x+1)}{x+1} \\ &= \lim_{x \rightarrow -1} x = -1 \neq h(-1) = 0 \end{aligned}$$

$$k(x) = \begin{cases} \frac{x^2+x-6}{x^2-x}, & x \neq 2 \\ -1, & x = 2 \end{cases}, \quad a = 2 \quad \times$$

$$k(2) = -1$$

$$\lim_{x \rightarrow 2} \frac{x^2+x-6}{x^2-x} = \frac{0}{2} = 0 \neq k(2) = -1$$

$$j(x) = |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}, \quad a = 0 \quad \checkmark$$

$$j(0) = 0$$

$$\lim_{x \rightarrow 0^-} j(x) = \lim_{x \rightarrow 0^-} -x = 0$$

$$\lim_{x \rightarrow 0^+} j(x) = \lim_{x \rightarrow 0^+} x = 0$$

$$\Rightarrow \lim_{x \rightarrow 0} j(x) = 0$$

Properties of Continuous Functions

1. The constant function $f(x) = c$ is continuous everywhere.
2. The identity function $f(x) = x$ is continuous everywhere.

If f and g are continuous at $x = a$, then

$[f(x)]^n$, where n is a real number, is continuous at $x = a$ whenever it is defined at that number

$f \pm g$ is continuous at $x = a$

fg is continuous at $x = a$

f/g is continuous at $x = a$ provided that $g(a) \neq 0$

Polynomial and Rational Functions

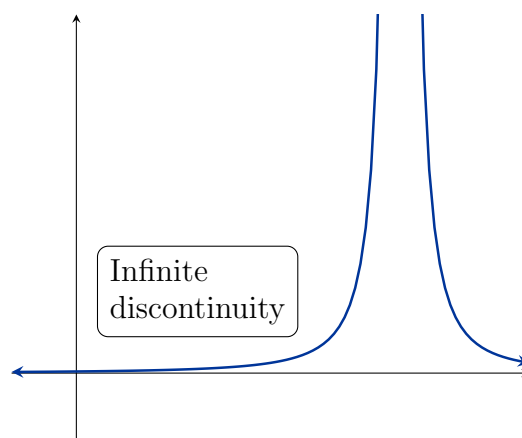
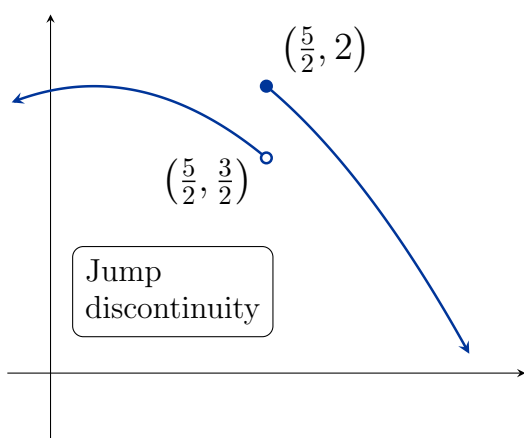
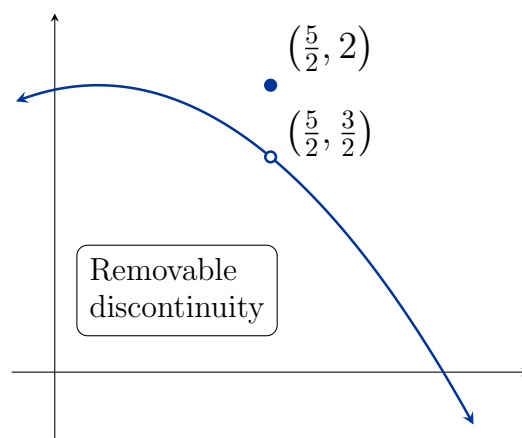
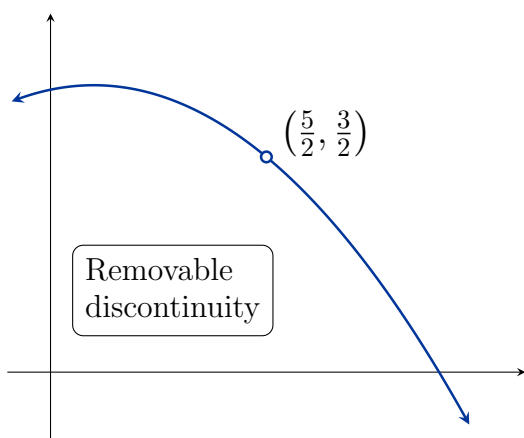
1. A polynomial function is continuous for all x .
2. A rational function (a function of the form $\frac{p}{q}$, where p and q are polynomials) is continuous for all x for which $q(x) \neq 0$.

Definition.

A **removable discontinuity** at $x = a$ is one that disappears when the function becomes continuous after defining $f(a) = \lim_{x \rightarrow a} f(x)$.

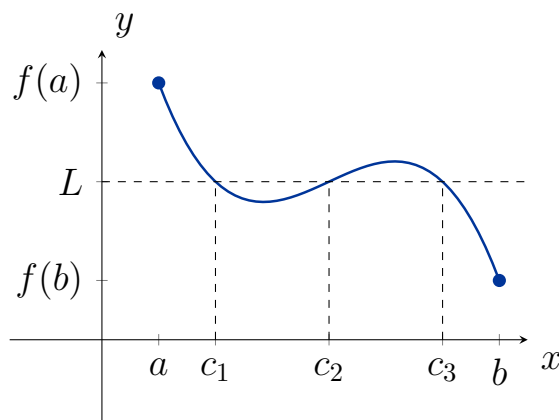
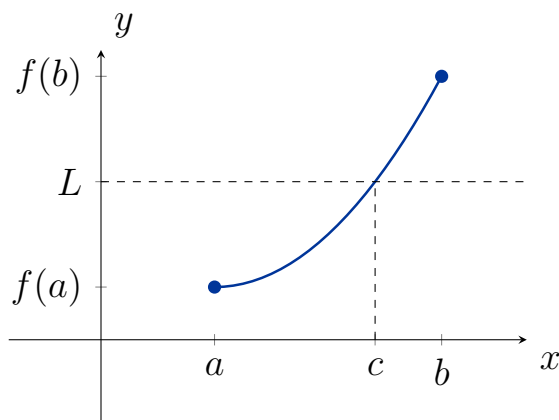
A **jump discontinuity** is one that occurs whenever $\lim_{x \rightarrow a^-} f(x)$ and $\lim_{x \rightarrow a^+} f(x)$ both exist, but $\lim_{x \rightarrow a^-} f(x) \neq \lim_{x \rightarrow a^+} f(x)$.

A **vertical discontinuity** occurs whenever $f(x)$ has a vertical asymptote.

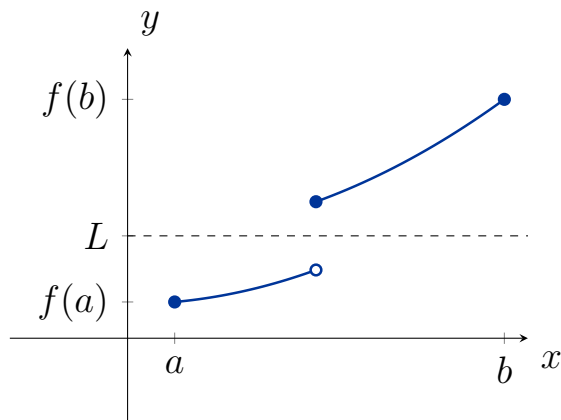


Theorem 4: Intermediate Value Theorem

Suppose f is continuous on the interval $[a, b]$ and L is a number strictly between $f(a)$ and $f(b)$. Then there exists at least one number c in (a, b) satisfying $f(c) = L$.



Note: It is important that the function be continuous on the interval $[a, b]$:

**Theorem 5: Existence of Zeros of a Continuous Function**

If f is a continuous function on a closed interval $[a, b]$, and if $f(a)$ and $f(b)$ have opposite signs, then there is at least one solution of the equation $f(x) = 0$ in the interval (a, b) .

Example. Check the conditions of the Intermediate Value Theorem to see if there exists a value c on the interval (a, b) such that the following equations hold: [Graph](#)

$$x^x - x^2 = \frac{1}{2} \quad \text{on } [0, 2]$$

$$x=0: 0^0 - 0^2 = 1$$

$$x=2: 2^2 - 2^2 = 0$$

Since $0 < \frac{1}{2} < 1$, there exists c such that $0 < c < 2$ and $f(c) = \frac{1}{2}$

$$\sqrt{x^4 + 25x^3 + 10} = 5 \quad \text{on } [0, 1]$$

$$x=0: \sqrt{0+0+10} = \sqrt{10} \approx 3.16$$

$$x=1: \sqrt{1+25+10} = \sqrt{36} = 6$$

Since $\sqrt{10} < 5 < 6$, there exists c such that $0 < c < 1$ and $f(c) = 5$

$$x + \sqrt{1 - x^2} = 0 \quad \text{on } [-1, 0]$$

$$x=-1: -1 + \sqrt{1-1} = -1$$

$$x=0: 0 + \sqrt{1-0} = 1$$

Since $-1 < 0 < 1$, there exists c such that $-1 < c < 0$ and $f(c) = 0$

$$\frac{x^2}{x^2 + 1} = 0 \quad \text{on } [-1, 1]$$

$$x=-1: \frac{(-1)^2}{(-1)^2 + 1} = \frac{1}{2}$$

$$x=1: \frac{(1)^2}{(1)^2 + 1} = \frac{1}{2}$$

Since the conditions of the IVT are not met, c such that $-1 < c < 1$ and $f(c) = 0$ may not exist (not guaranteed)

Note: $x=0$ is a root.

Example. Consider the function

$$f(x) = \frac{x+1}{x-1}$$

on the interval $[0, 2]$. Does there exist a c on the interval $[0, 2]$ such that $f(c) = 1$?

$$\left. \begin{array}{l} f(0) = -1 \\ f(2) = 3 \end{array} \right\} -1 < 1 < 3$$

Since $f(x)$ is discontinuous
at $x=1$, we cannot apply
the IVT.

